Physical AI

A Comprehensive Framework for Launching AI-Powered Physical Products

14 Days	200+	$20K+
First Prototype	Factory Partners	Starting Investment

McGlobal — Your Outsourced Physical AI R&D; Department

www.mcglobal.cc | build@mcglobal.cc

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Region	Market Size	Key Segments	Regulatory Complexity
North America	$28B	Consumer AI, Healthcare	High (FCC, FDA, CPSIA)
Europe	$18B	Industrial AI, Smart Home	Very High (CE, GDPR)
Asia-Pacific	$25B	Manufacturing, Robotics	Medium
Rest of World	$7B	Emerging Consumer AI	Low-Medium

Competitive Landscape

Key Trends in 2026

4. Regulatory Requirements

United States — FCC Requirements

Certification Type	When Required	Timeline	Cost Range
FCC SDoC (Self-Declaration)	Non-wireless devices	2-4 weeks	$2,000-$5,000
FCC Certification	Wireless/radio products	4-8 weeks	$5,000-$15,000
FCC ID Labeling	All FCC products	N/A	$500-$1,000

Children's Products — CPSIA

Data Privacy — COPPA & State Laws

Canada — CCPSA

The Compliance-First Approach

5. Supply Chain Overview

Why Shenzhen for AI Hardware?

Typical Supply Chain Timeline

Phase	Duration	Key Activities
Component Sourcing	3-5 days	BOM finalization, supplier identification
PCB Manufacturing	3-5 days	Board fabrication, component placement
Mechanical Prototype	5-7 days	3D printing, injection molding samples
Assembly & Integration	2-3 days	PCB integration, sensor calibration
Testing & QC	2-3 days	Functionality testing, compliance prep
Total	**14-21 days**	**Mass-production-ready prototype**

Minimum Order Quantities (MOQ)

Production Phase	Typical MOQ	Lead Time
Prototype/Sample	1-5 units	14-21 days
Small Batch	100-500 units	3-4 weeks
Trial Production	500-1,000 units	4-6 weeks

Mass Production	5,000+ units	6-12 weeks

6. The 14-Day Launch Framework

Week 1: Discovery & Build

Day	Activity	Deliverable
1	Discovery Call & Brief Lock	Signed project brief, initial requirements
2-3	Industrial Design Concepts	3 concept renders, design direction selected
4-5	AI Architecture & BOM Draft	Technical spec document, preliminary BOM
6-7	PCB Layout & Mechanical Prep	Board files, initial housing prototypes

Week 2: Integration & First Prototype

Day	Activity	Deliverable
8-9	Component Assembly	Assembled PCB, sensor integration
10-11	Firmware Flash & AI Calibration	Working firmware, AI model testing
12-13	Mechanical Assembly & Testing	Physical prototype, initial QC report
14	**Demo Day — First Prototype Delivery**	**Working prototype, test results, next steps**

What Happens After Day 14?

7. Cost Breakdown

Phase 1: Prototype (Months 1-2)

Item	Low Estimate	High Estimate
Industrial Design	$3,000	$8,000
PCB Development	$2,000	$6,000
Mechanical Prototype	$2,000	$5,000
AI/ML Integration	$3,000	$10,000
Component Procurement	$2,000	$8,000
Assembly & Labor	$1,500	$4,000
Testing Equipment	$1,000	$3,000
Travel (if applicable)	$0	$5,000
TOTAL	**$14,500**	**$49,000**

Phase 2: Certification (Months 2-4)

Phase 3: Production (Month 3+)

Total Realistic Investment

Phase	Timeline	Investment Range
Prototype	Month 1-2	$15,000 - $50,000
Certification	Month 2-4	$18,000 - $60,000
First Production	Month 3-5	$45,000 - $150,000
To Market (Total)	**Month 4-6**	**$80,000 - $260,000**

8. Common Pitfalls & How to Avoid Them

9. Next Steps

Step 1: Define Your Product Scope

Step 2: Assess Your Budget Reality

Step 3: Get Expert Guidance

Ready to Start?
Book your free consultation at mcglobal.cc
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About McGlobal